

FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10: 3.2 CURRENT ELECTRICITY

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This Final Explanation brings together everything you have learned in Lessons 1 to 12. Answer each question in the space under it, using your own words, the evidence and measurements from our bench demos (9 V pack, three identical 9 V, 2.7 W, 30 Ω bulbs) and the Nairobi classroom. Show every calculation step with units. Use the terms 'in series', 'in parallel', 'potential difference (p.d.)' and 'e.m.f.' correctly. Remember that an ammeter is connected in series and a voltmeter in parallel. Your final answer should connect back to the driving question: Why do the other tubes stay lit when one blows out, and why do bulbs connected in series glow dimmer than identical bulbs connected in parallel to the same battery?

Part 1: Why the other tubes stay lit

In the Nairobi evening prep class, four fluorescent tubes share the 240 V mains and one tube blows. (a) How are the tubes connected, and what happens to the other three? (b) Using what you learned about closed loops, p.d. and current, explain why the other three tubes are not affected. (c) Each tube is 36 W on 240 V. Calculate the current in one tube, the current for all four tubes, and the current from the supply after one tube blows. (d) Compare this with an old Christmas-light string in which one bulb fails: what happens, and why?

Write your answer here:

Part 2: Why the series bulbs are dim

Mr. Otieno connected three identical 9 V, 2.7 W, $30\ \Omega$ bulbs in series to the 9 V battery pack. Wanjiru asked why they glow dim. (a) Calculate the total resistance of the circuit. (b) Calculate the current in the circuit and say what an ammeter would read at different points in the loop. (c) Calculate the p.d. across each bulb and show how the p.d.s relate to the e.m.f. of the pack. (d) Calculate the power in each bulb and use it to explain why they are dim. (e) What happens if one bulb is removed, and why?

Write your answer here:

Part 3: Why the parallel bulbs are bright

Now the three identical bulbs are connected in parallel to the same 9 V pack. (a) What p.d. is across each bulb, and what current flows in each branch? (b) What current comes from the battery, and what is the effective resistance of the circuit? Show it using $1/R = 1/R_1 + 1/R_2 + 1/R_3$. (c) Calculate the power in each bulb and compare it with a series bulb. (d) One bulb is removed. What are the p.d. and current in each remaining bulb, and what is the battery current now? (e) Kipchirchir predicted that the battery is used up faster in parallel. Is he right? Use your numbers.

Write your answer here:

Part 4: Energy, cost and safe design

The school wants to light the Nairobi classroom with the four 36 W tubes in parallel on 240 V mains. (a) Calculate the energy used by the four tubes in 5 hours in kWh and the cost at KSh 25 per kWh. (b) A 60 W filament bulb and a 9 W LED are each used for 5 hours. Compare the energy used and the cost of each. (c) Achieng's family in Kisumu has a 12 V solar home system. Say how its lights should be connected so that one failed bulb does not put out the others, and why. (d) A kettle (3000 W) and a heater (2000 W) are plugged into one 240 V socket with a 13 A fuse. Calculate what happens and explain why. (e) Describe the safety devices that your classroom circuit should have.

Write your answer here:

Part 5: Final explanation and model revision

Write a complete final explanation that answers the driving question: Why do the other tubes stay lit when one blows out, and why do bulbs connected in series glow dimmer than identical bulbs connected in parallel to the same battery? Use evidence from at least three of our lessons (for example the current and p.d. measurements, Ohm's law, and resistance in series and in parallel). Then state one idea in your Lesson 1 model that you now want to change, and say how you would change it. Finally, say how this helps you answer the KICD questions: how do we determine the relationship between current, p.d. and resistance, and how is current electricity applied in day-to-day life?

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence and data	Uses the exact values from the demos and classroom (90 Ω , 10 Ω , 0.10 A, 0.30 A, 3.0 V, 9.0 V, 0.30 W, 2.7 W, 0.15 A, 0.60 A, 0.45 A) accurately and links them to claims.	Uses most of the correct values but some are missing or not linked clearly to the claim.	Uses few or incorrect values, or gives claims without evidence.
Scientific reasoning about series and parallel circuits	Clearly explains that series bulbs share the p.d. and a single current through 90 Ω , while parallel branches each get the full p.d. and independent current, and links this to why one blown tube does not affect the others.	Explains the series and parallel difference correctly but with limited reasoning about p.d., current or resistance.	Describes what happens without a correct reason, or shows misconceptions such as bulbs using up current.
Calculations and units	All calculations ($V = IR$, resistance in series and parallel, $P = IV$, $E = Pt$, cost) are set out step by step with correct units and answers.	Calculations mostly correct with minor errors in units or steps.	Calculations incomplete or incorrect, or missing units.
Application and safety	Correctly explains the fuse blowing at 20.8 A against 13 A, the roles of the fuse or breaker and earthing, and the cost saving of an LED, in Kenyan contexts.	Explains most safety and cost ideas correctly with some gaps.	Safety and cost ideas are unclear or missing.

Model revision and communication	Compares the first model with the final model, names a specific change, links to both KICD questions, and uses scientific terms (in series, in parallel, p.d., e.m.f.) correctly.	Describes a change in the model and uses most terms correctly.	Does not describe a model change or uses terms incorrectly.
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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